

Digital Signal Processing Laboratory

Paper Code: EC 593

Assignment -1

Familiarization with OCTAVE command and function

1. Solve $[3^2 - ((5+4) / 2) + 6*3]$ using OCTAVE
2. Determine the value of $x^2 + 5x + 2$ when $x=2$.
3. Create one dimensional vector Z of elements 2,4,6,8.
4. Create a vector X of values running from 20 to 29.
5. Change the vector X from a row vector to a column vector.
6. Square the elements of the vector X.
7. Create a column vector whose values running from 1 to 7.
8. Create a complex row vector E from the array Z. Then create a column vector A that is complex conjugate transpose of vector E. Again create a column vector F that is transpose of E.
9. Create two vectors X and Y. The value of vector X running from 1 to 9 and the value of vector Y running from 2 to 10. After that multiplying the element of X by the corresponding element of vector Y.
10. Construct two arrays of dimensions 3 by 3 G and H and obtain the value of G-H.
11. Create an array G of dimension 3 by 3. Subtract 2 from each element.
12. Create an array of any dimension that contains all one.
13. Create an array of any dimension containing all zeros. Determine the dimension of the array.
14. Create an Identity matrix whose dimension is (2,5).
15. Construct a simple array A of [1 2 3 4]. Then convert it into diagonal arrays.
16. Create an array $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ Set the element in 2 row 6 column to one then set the 4 column of vector A to 5.
17. Create C is a 2 by 2 array then converts it into a row vector.

Problems On Two Dimensional Graphics

1. Plot the function given as $y = \sin(x)$ where $0 < x < 2\pi$ considering 30 points between 0 and 2π .
2. Plot more than one curve: $\sin(x)$ and $\cos(x)$ on the same plot.
3. Plot $\sin(x)$, $\cos(x)$ on the same plot using a color, a marker and a line style.
4. Plot $\sin(x)$, $\cos(x)$ on the same plot using grid lines, labels.
5. Plot $\sin(x)$, $\cos(x)$ on the same plot using hold on command.
6. Plot $\sin(x)$, $\cos(x)$, $2 * \sin(x)$, $* \cos(x)$, $\sin(x) ./ \cos(x)$ in one window using subplot command.